Qian Yang

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EDUCATION

Mila - Quebec AI Institute & Université de Montréal 2023.09 – Present
Ph.D. in Computer Science | CGPA: 4.3/4.3 | Advisor: Prof. Aishwarya Agrawal
Research: Efficiency and Reliability of Multi-modal Large Language Models

Harbin Institute of Technology, Shenzhen 2020.09 – 2023.03
MSc in Computer Science and Technology | Advisor: Prof. 户保田教授
Research: Fine-grained Alignment for Explainable Multi-modal Inference

University of Electronic Science and Technology of China 2016.09 – 2020.06
BEng in Computer Science and Technology | CGPA: 3.73/4.0 (Top 10%)

RESEARCH HIGHLIGHTS & CORE COMPETENCIES

- **Efficient Multimodal Training:** Focused on **Data-centric AI** for VLMs. Developed the SAIL framework (CVPR'25 Highlight) achieving CLIP-level performance with **6% data**, and the PROGRESS framework (CVPR'26) for high-efficiency instruction tuning.
- **VLM Reliability & Factuality:** Specialized in quantifying and mitigating hallucinations in Large Models. Developed task-decomposition consistency metrics (EMNLP'24) and execution-based code hallucination benchmarks (AAAI'25).
- **Distributed RL Systems:** Proficient in large-scale Reinforcement Learning using **verl** and **DeepSpeed**. Experienced in designing Fine-grained Reward Models and optimizing training efficiency in multi-node GPU clusters.

PUBLICATIONS

- **Qian Yang***, Shivam Chandhok*, ..., Leonid Sigal, Aishwarya Agrawal. Learning What Matters: Prioritized Concept Learning via Relative Error-driven Sample Selection. CVPR 2026.
  – **Efficient Tuning:** Developed a dynamic sample selection framework for VLM instruction tuning. By identifying high-value samples via relative error, the method **reduces data and compute costs by 80%** while maintaining SOTA performance across multimodal benchmarks.
- Le Zhang, **Qian Yang**, Aishwarya Agrawal. Assessing and Learning Alignment of Unimodal Vision and Language Models. CVPR 2025 (***Highlight***).
  – **Low-Resource Alignment:** Introduced the SAIL framework, matching CLIP's zero-shot performance using **only 5 hours of single-GPU training** and 6% of the standard paired dataset. Demonstrated the extreme efficiency of unimodal-to-multimodal alignment.
- **Qian Yang**, Weixiang Yan, Aishwarya Agrawal. Enhancing Multi-Agent Multi-Modal Collaboration with Fine-Grained Reward Modeling. *NeurIPS 2024 Workshop on AFM.*
  – **RL Systems:** Built a multi-agent RL pipeline using the verl framework. Implemented a Process Reward Model to address sparse rewards in complex collaboration tasks.
- Yuchen Tian, Weixiang Yan, **Qian Yang**, ..., Dawn Song. CodeHalu: Investigating Code Hallucinations in LLMs via Execution-based Verification. AAAI 2025.
  – **Benchmarking:** Developed an automated execution-based verification pipeline for code hallucination detection. Evaluated 17 mainstream LLMs using the 8.8k sample CodeHalu benchmark to identify

systematic reasoning failures.

- **Qian Yang**, Weixiang Yan, Aishwarya Agrawal. Decompose and Compare Consistency: Measuring VLMs' Answer Reliability via Task-Decomposition Consistency Comparison. EMNLP 2024.
  – **Reliability:** Designed a task-agnostic evaluation metric for VLM reliability. By comparing consistency between direct answers and decomposed sub-questions, the method effectively identifies overconfident hallucinations across 6 major tasks.
- Le Zhang, Yihong Wu, **Qian Yang**, Jianyun Nie. Exploring the Best Practices of Query Expansion with Large Language Models. EMNLP Findings 2024.
- **Qian Yang**, ..., Min Zhang. Enhancing Multi-modal and Multi-hop Question Answering via Structured Knowledge and Unified Retrieval-Generation. ACM MM 2023.
- **Qian Yang**, Yunxin Li, ..., Min Zhang. Chunk-aware Alignment and Lexical Constraint for Visual Entailment with Natural Language Explanations. ACM MM 2022.
- Yunxin Li, **Qian Yang**, ..., Lin Ma. Fast and Robust Online Handwritten Chinese Character Recognition with Deep Spatial & Contextual Information Fusion Network. TMM 2022.

Preprints

- **Qian Yang**, ..., Aishwarya Agrawal. How and What to Imagine? Visual Thinking in Unified Multimodal Models for Cross-View Spatial Reasoning (Under Review) 2026
- Kanishk Jain, **Qian Yang**, ..., Aishwarya Agrawal. Discovering Failure Modes in Vision-Language Models using RL (Under Review) 2026

ACADEMIC INTERNSHIPS

Alibaba DAMO Academy, Hangzhou, China *Advisors: Dr. Wen Wang, Dr. Qian Chen*
Multimodal Retrieval-Augmented Generation (RAG) & Knowledge Alignment *2022.05 – 2022.10*

- **Multimodal RAG Architecture:** Developed a unified **Retrieval-Generation** framework to integrate intermediate cross-modal reasoning steps. Effectively mitigated information gaps in multi-hop question answering, significantly enhancing the model's reasoning depth.
- **Entity-Centric Fusion:** Architected an entity-based alignment model utilizing structured knowledge graphs to bridge the semantic gap between visual features and textual concepts. Solved the "long-tail entity" recognition challenge in multimodal environments.
- **Impact:** Successfully implemented the pipeline within the team's research stack; findings were published as a full paper in **ACM Multimedia 2023**.

AWARDS AND SCHOLARSHIPS

FRQNT (Fonds de recherche du Québec), Québec, Canada (25,000 CAD/year)	2026 – 2029
AI Scholarship, University of Montreal (10,000 CAD)	2026
Women in Machine Learning (WiML), MILA (7,000 CAD)	2025
Student Success Scholarship, University of Montreal (5,000 CAD)	2025
DIRO Excellence Scholarship, University of Montreal (5,000 CAD)	2024 – 2025
Professor Cho Diversity Scholarship, MILA (1,500 CAD)	2025
National Encouragement Scholarship (**Top 10%**, National Level Award)	2019
First Prize Academic Scholarship, UESTC (**Top 20%**)	2016 – 2020

ACADEMIC SERVICE & ACTIVITIES

Workshop Organization

- **Vision Language Models For All (VLMs4All)** | *Organizer, CVPR 2025 Workshop*
  – Co-organizing a venue focused on Building Geo-Diverse and Culturally Aware VLMs; driving discussions on model robustness and global generalization at scale.

Academic Reviewing

- **CVPR 2025 Outstanding Reviewer** (Top-tier recognition for review quality)
- **Program Committee / Reviewer:** CVPR 2024-26, ECCV 2024, AAAI 2024, ACM Multimedia 2023-24, COLING 2022

Teaching Experience

- **Representation Learning** (IFT6135, Fall 2024) – *University of Montreal*
  – Graduate-level TA for a flagship Deep Learning course; responsible for designing assignments, leading lab sessions, and providing technical guidance on SOTA architectures.
- **Mathematical Logic / Intro to Algorithms** (2020-2021) – *HIT, Shenzhen*

TECHNICAL SKILLS

- **Programming:** Python (Expert), C/C++, Shell Scripting (Bash)
- **DL & Distributed Systems:** PyTorch, verl (Ray + vLLM), DeepSpeed (ZeRO 1/2/3), FlashAttention
- **Infrastructure & Tools:** SLURM Cluster Management, Linux (Ubuntu/CentOS), Git, Docker, WandB
- **Languages:** English (TOEFL: 99 — R:26, L:25, S:22, W:26), Mandarin (Native)